CSC-343: Midterm Exam

(NLP Before LLMs Review Questions)

- **Show all work** for full credit. Answers without supporting steps or reasoning will receive little or no credit.
- Present your work in a **clear, logical, and organized manner**. Do not scatter steps with arrows or disconnected notes.
- **Clearly indicate your final answers** (e.g., box or highlight them).
- For proofs or problems, state all assumptions explicitly and justify each step.
- Only non-graphing calculators are permitted. No other electronic devices (including cell phones) are allowed.
- Use a pencil or erasable pen for your responses.
- For questions requiring responses that are not computational, you must use complete sentences.
- Any extraneous commentary about exam content will result in a reduced professionalism grade. If you have concerns, ask clarifying questions instead.
- Scratch paper is for your use only. It will be collected but not reviewed. Transfer all final work to this exam.
- Your responses should reflect the notation and techniques presented in class. While alternative conventions exist, consistency with course material is required for full credit.

By signing below, I certify that the work on this exam is my own without the use of external resources.

_____ _____

Printed Name Signature

Short Answer

1. What is the fundamental difference between symbolic/rule-based NLP systems and statistical NLP systems? Explain why the statistical approach is considered more robust.

2. Describe the data sparsity problem that arises when trying to estimate $P(w_n \mid w_1, w_2, \dots, w_{n-1})$ for long sequences. Why can't we simply memorize probabilities for all possible word sequences?

3. Explain the distributional hypothesis "You shall know a word by the company it keeps". How does this principle justify using word co-occurrence patterns to learn semantic representations?

4. Why do we use log-likelihood instead of raw likelihood when training language models and word embeddings? What numerical problem does this solve?

5. Describe three limitations of simple word embeddings like Word2Vec.

6. Explain the relationship between perplexity and cross-entropy. How do you convert from one to the other?

True or **False**

1. A bigram language model makes the assumption that the probability of a word depends only on the single word that precedes it.

2. In distributional semantics, words that frequently appear together in sentences (like "king" and "throne") are expected to have more similar vector representations than words that appear in similar contexts (like "doctor" and "nurse").

3. Add-1 (Laplace) smoothing tends to overestimate the probability of unseen events more than add-k smoothing with $k < 1$.

4. A unigram language model can capture grammatical structure and word order dependencies in generated text.

5. Perplexity of 100 indicates that the language model is equally uncertain between 100 possible next words on average.

6. In the Skip-gram model, each word is represented by two vectors: one when it serves as a center word and one when it serves as a context word.

7. Cosine similarity between two word vectors can range from -1 to 1, where values close to 1 indicate similar meanings.

8. One-hot vector representations inherently capture semantic similarity between words because similar words have similar binary patterns.

9. The Markov assumption in n-gram models exists primarily to address the data sparsity problem that arises as context length increases.

Other Questions

1. N-gram Probability Calculation: Given the following bigram counts from a corpus:

```
Count("the cat"): 15

Count("the"): 500

Count("cat sat"): 8

Count("cat"): 50

Count("sat"): 80
```

Calculate each of the following:

(a) `P("cat" | "the")` using maximum likelihood estimation. [0.03]
(b) `P("sat" | "cat")` using maximum likelihood estimation. [0.16]
(c) `P("the cat sat")` using the bigram model. [0.0048]

2. Perplexity Comparison: Two language models are evaluated on the same test corpus:

Model A: cross-entropy = 4.5 bits/word

Model B: cross-entropy = 6.2 bits/word

(a) Calculate the perplexity for each model.
(b) Which model is better and why?
(c) Interpret the perplexity values: what does Model A's perplexity tell us about the average number of equally likely choices the model faces?

3. Consider a vocabulary of size $|V| = 10{,}000$. You observe:

```
Count("quantum", "physics") = 3

Count("quantum") = 100
```

Calculate `P("physics" | "quantum")` using:

(a) Maximum likelihood (no smoothing) [0.03]
(b) Add-1 (Laplace) smoothing [0.000396]
(c) Add-0.1 smoothing [0.002818]

4. A bigram language model assigns the following probabilities to a test sentence "the dog barked":

```
P("the" | <START>) = 0.1

P("dog" | "the") = 0.02

P("barked" | "dog") = 0.05

P(<END> | "barked") = 0.4
```

Calculate:

(a) The probability of the entire sentence. [0.00004]
(b) The cross-entropy in $\frac{bits}{word}$. [3.65]
(c) The perplexity. [12.55]

5. Consider the sentence "the quick brown fox jumps" with window size 2.

(a) List all (center word, context word) pairs that would be generated for training. [14 pairs]
(b) If we instead used window size 1, how many fewer training pairs would we get? [8 pairs]
(c) Briefly explain how window size affects what semantic relationships are captured.